

ENMA301 Homework #4

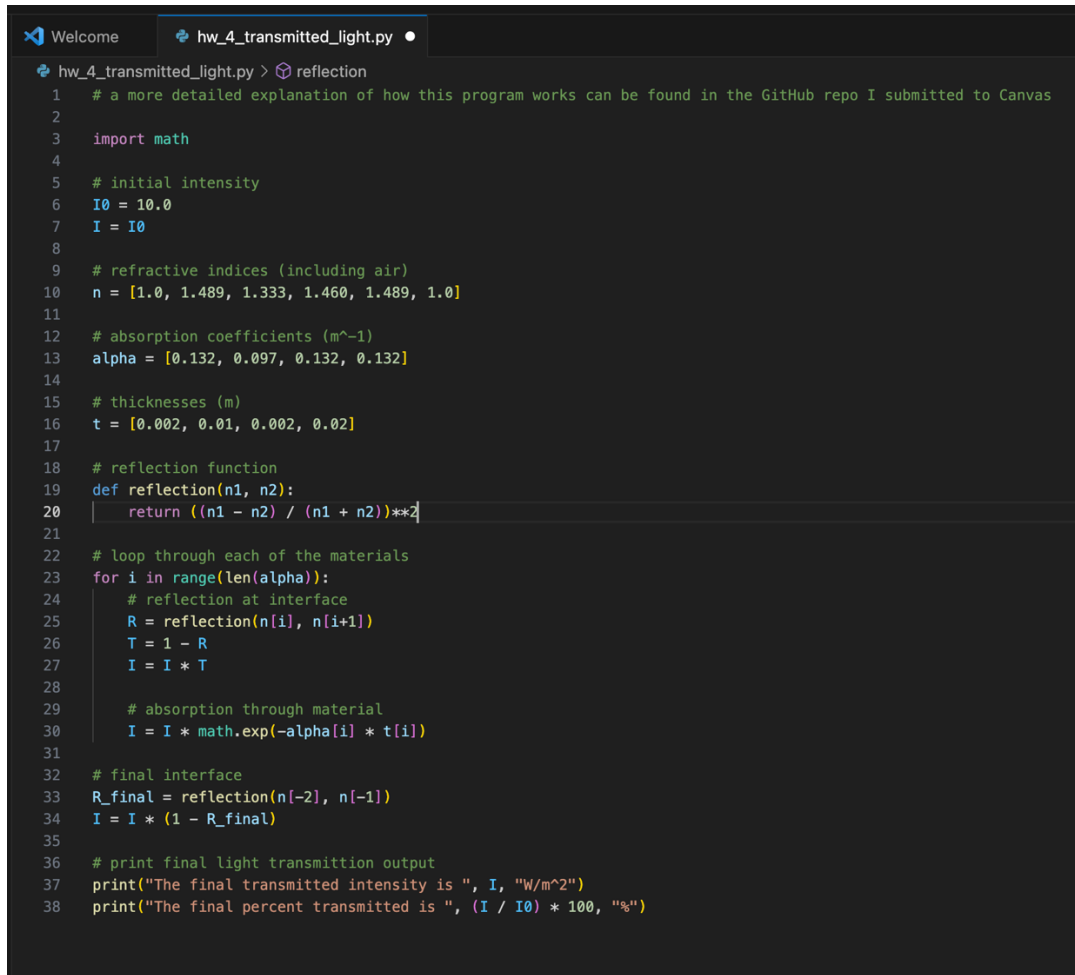
1. Explanation of calculating transmitted light

- Light loses its intensity due to reflection at the interfaces and because it gets absorbed into each material
- At each boundary between two materials, $R = [(n_1 - n_2) / (n_1 + n_2)]^2$ where n_1 and n_2 are the refractive indices and R is the fraction of light reflected
 - The transmitted portion is $T = 1 - R$ at each interface
- Light intensity decreases exponentially according to the Beer Lambert law ($I = I_0 e^{-\alpha x}$)
- So to calculate this, I will start with the initial intensity of $I_0 = 10 \text{ W/m}^2$, then for each layer, apply the reflection loss at interface and absorption through the material
- Repeat for all layers (Air $\rightarrow$ PMMA $\rightarrow$ Water $\rightarrow$ Polymer $\rightarrow$ PMMA $\rightarrow$ Air)
- The final intensity is the product of all transmission/absorption effects

2. Thought process

- I represented the system as layers in a specific order as seen above
- I then stored the properties in lists (refractive indices, absorption coeff., thicknesses)
- In the program, I defined a function that calculated reflection coefficient between two materials, then looped that so it would basically iterate through each material layer and apply the reflection/absorption sequentially
- For the computation, I set initial intensity to 10 as mentioned above, then for each layer, compute the reflection loss, multiply intensity by transmission factor, and then apply the absorption which is exponential
 - After the loop is done, I applied the final interface loss from PMMA to air
 - The output is the final intensity and % transmitted

3. Program screenshots

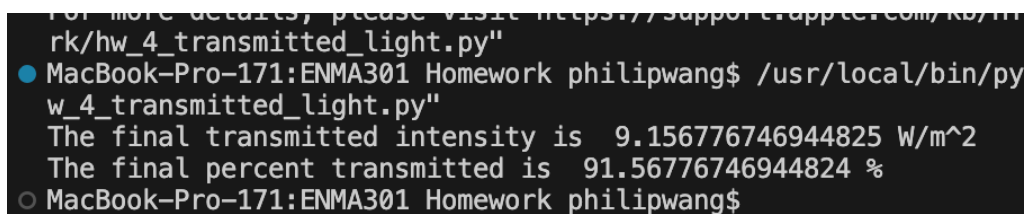


```
1 # a more detailed explanation of how this program works can be found in the GitHub repo I submitted to Canvas
2
3 import math
4
5 # initial intensity
6 I0 = 10.0
7 I = I0
8
9 # refractive indices (including air)
10 n = [1.0, 1.489, 1.333, 1.460, 1.489, 1.0]
11
12 # absorption coefficients (m^-1)
13 alpha = [0.132, 0.097, 0.132, 0.132]
14
15 # thicknesses (m)
16 t = [0.002, 0.01, 0.002, 0.02]
17
18 # reflection function
19 def reflection(n1, n2):
20     return ((n1 - n2) / (n1 + n2))**2
21
22 # loop through each of the materials
23 for i in range(len(alpha)):
24     # reflection at interface
25     R = reflection(n[i], n[i+1])
26     T = 1 - R
27     I = I * T
28
29     # absorption through material
30     I = I * math.exp(-alpha[i] * t[i])
31
32 # final interface
33 R_final = reflection(n[-2], n[-1])
34 I = I * (1 - R_final)
35
36 # print final light transmission output
37 print("The final transmitted intensity is ", I, "W/m^2")
38 print("The final percent transmitted is ", (I / I0) * 100, "%")
```

4. Compiled program

GitHub repo link: https://github.com/phichawa/ENMA_301_HW4_LightTransmission.git

5. Results screenshot



```
For more details, please visit https://support.apple.com/kb/nr/hw_4_transmitted_light.py"
MacBook-Pro-171:ENMA301 Homework philipwang$ ./usr/local/bin/pyw_4_transmitted_light.py
The final transmitted intensity is 9.156776746944825 W/m^2
The final percent transmitted is 91.56776746944824 %
MacBook-Pro-171:ENMA301 Homework philipwang$
```

6. If the polymer swells

- Thickness will increase from the Beer Lambert law ($I = I_0 e^{-\alpha t}$)
- As t increases the exponent becomes MORE negative, so the exponential term e ultimately decreases
- This means that more light is absorbed in the polymer layer, and since intensity is reduced, the final transmitted intensity decreases

7. If the polymer dissolves

- Use the BL law and similar logic again
- As thickness decreases because of dissolving, the exponent becomes LESS negative, so the entire exponential term increases
- Less light is absorbed in the layer, and so since less intensity is lost, the final transmitted intensity increases